Jeonghyeon Kim

M.S. Student in Mechanical Engineering, Korea University

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Education

Korea University *Mar. 2025 – Feb. 2027 (Expected)*
M.S. in Mechanical Engineering Seoul, Republic of Korea

Advisor: Prof. Shinsuk Park

GPA: 3.78/4.0 · Research interests: control-oriented dynamics modeling and whole-body control

Korea University *Mar. 2019 – Feb. 2025*
B.S. in Mechanical Engineering Seoul, Republic of Korea

GPA: 3.48/4.0

Research Experience

Balance-Aware Whole-Body Control for Humanoid Standing Manipulation
Human-Machine Systems Lab, Korea University *Mar. 2025 – Present*

Formulated a **Balance Demand Descriptor (BDD)** based on centroidal dynamics to represent balance demands induced by upper-body motion, payloads, and external interaction. Integrated BDD into a reinforcement-learning locomotion policy to distinguish demands manageable within the current stance from those requiring corrective stepping.

Visual Perception for Quadrupedal Object Interaction
Human-Machine Systems Lab, Korea University *Mar. 2025 – Nov. 2025*

Developed a dual-camera perception system to track a ball and moving ArUco markers from fixed global and onboard views. Calibrated the cameras and transformed observations into a common coordinate frame to maintain target localization during robot motion and partial occlusion.

Real-Time Architecture for a Generative-AI-Based Mobile Manipulator
Human-Machine Systems Lab, Korea University *Mar. 2025 – Aug. 2026*

Developed a ROS 2 communication architecture connecting simulation, an LLM, a user interface, and a physical mobile manipulator. Implemented edge-based robot data collection with automatic cloud synchronization while preserving the robot's real-time control cycle.

Publications & Manuscripts

Accepted

H.-Y. Ju, **J.-H. Kim**, and S.-S. Park, "A Study on Improving Sim-to-Real Accuracy through Dynamics Parameter Tuning in Reinforcement Learning-based Mapless Navigation," *Journal of Institute of Control, Robotics and Systems*, accepted, 2026.

Submitted Manuscripts

J.-H. Kim and S.-S. Park, "Feasible Is Not Faithful: Intent-Preserving Constraint Realization for Contact Tasks on Unknown Surfaces," *IEEE Robotics and Automation Letters*, submitted, Nov. 2026. Preprint on arXiv.

Manuscripts in Preparation

J.-H. Kim and S.-S. Park, "Balance Demand Descriptor: A Physics-Informed Representation of Whole-Body Dynamics for Humanoid Standing Manipulation," manuscript in preparation.

J.-H. Kim, J. Kim, and S.-S. Park, "Hold Before You Step: IMU-Resolved Disturbance Estimation for In-Place Push Recovery on Force-Sensorless Humanoids," manuscript in preparation. Preprint on arXiv.

Conference Presentations

J.-H. Kim and S.-S. Park, "Obstacle Detection Using ORB-SLAM and UV-Disparity with a Low-Cost Quadruped Robot," *KSPE 2025 Spring Conference*, Jeju, Republic of Korea, May 2025.

Engineering Design Projects

Quadruped Robot Obstacle Avoidance *Mar. 2024 – Jan. 2025*

Led the development of an autonomous obstacle detection and avoidance system for the DOGZILLA quadruped robot. Integrated ORB-SLAM with UV-disparity-based obstacle detection, earning the President's Award in the Korea University Creative Challenger Program.

Ladder-Crossing Robot *Mar. 2023 – Sep. 2023*

Designed and prototyped a mobile robot capable of traversing ladder-like structures. Developed the crossing mechanism and Arduino-based motion control, earning a Bronze Award at the Creative Comprehensive Design Competition.

Solar-Tracking Parasol *Sep. 2023 – Nov. 2023*

Designed a linkage mechanism and tracking controller that automatically adjusts the parasol's orientation with changes in incident-light direction, receiving the Design Award at the Korea University Capstone Design Fair.

Honors and Awards

Master's Research Encouragement Grant, National Research Foundation of Korea *2025*
President's Award, Creative Challenger Program, Korea University *Jan. 2025*
Bronze Award, Creative Comprehensive Design Competition, Korea University *Sep. 2023*
Design Award, Capstone Design Fair, Korea University *Nov. 2023*

Teaching Experience

Teaching Assistant, Korea University, Seoul, Republic of Korea

Robotics	Spring 2026
Computer-Aided Design	Fall 2025
Dynamics (English-taught course)	Summer 2025
Kinematics of Mechanisms	Spring 2025

Technical Skills

Programming	Python, C/C++, MATLAB
Robotics	ROS 2, Pinocchio, URDF/XACRO
Simulation	Isaac Gym, Isaac Sim, MuJoCo, Gazebo
Learning & Optimization	PyTorch, reinforcement learning, CVXPY, CasADi, trajectory optimization
Perception	OpenCV, ORB-SLAM
Engineering Tools	Git, SolidWorks, Creo, NX, Arduino
Operating Systems	Linux/Ubuntu